

BANGALORE NORTH UNIVERSITY



Project Report on

LAPTOP SELECTION AND PRICE PREDICTION

**Submitted in Partial Fulfilment of the Requirements of Bachelor of
Computer Application Degree of Bengaluru North University**

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NEW HORIZON COLLEGE

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NEW HORIZON COLLEGE

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CERTIFICATE

This is to certify that the project titled — **LAPTOP SELECTION AND PRICE PREDICTION** has been satisfactorily completed by **SADHANA R**, Reg No. U19KU23S0151, in partial fulfillment of the requirements for Project Lab, for the VI Semester BCA course during the academic semester from 2025 - 2026 as prescribed by Bengaluru North University.

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STUDENT DECLARATION

I, **SADHANA R** hereby declare that the project report entitled “**LAPTOP SELECTION AND PRICE PREDICTION**” was prepared by me during the year 2025 - 2026 and was submitted to the Bengaluru North University, in partial fulfillment of the requirements for the award of the Degree in **Bachelor of Computer Applications**. I also declare that this project report is original and genuine and it has not been submitted to any other University for the award of any degree, diploma or other similar titles or purposes.

Date:

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SADHANA R

[U19KU23S0151]

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ABSTRACT

In the modern digital era, laptops have become an important part of daily life for education, office work, business, gaming, and entertainment purposes. Due to the availability of numerous laptop brands and models in the market, selecting the best laptop according to user requirements and budget has become a challenging task. This project, titled **“Laptop Selection and Price Prediction”**, aims to develop a machine learning-based system that predicts the price of laptops based on their specifications and features.

The dataset used in this project contains information about various laptops, including processor type, RAM, storage capacity, operating system, display size, ratings, reviews, and price. The dataset is collected from online platforms and preprocessed using techniques such as handling missing values, removing duplicates, feature extraction, encoding categorical data, and data scaling. Exploratory Data Analysis (EDA) and data visualization are also performed to understand the relationship between different laptop features and their prices.

Several machine learning regression algorithms such as Linear Regression, Decision Tree Regression, Random Forest Regression, and Support Vector Regression (SVR) are implemented to predict laptop prices. The performance of these models is evaluated using metrics like Root Mean Square Error (RMSE) and R2 Score to identify the most accurate prediction model. Among these algorithms, Random Forest Regression provides better prediction accuracy and performance compared to other models.

The project is implemented using Python programming language along with libraries such as NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn. The developed system helps users make better purchasing decisions by estimating laptop prices based on technical specifications. This project demonstrates how machine learning techniques can be effectively applied in real-world applications for accurate prediction, data analysis, and smart decision-making.

CHAPTER 1: INTRODUCTION

1.1 General Introduction

In today's digital era, laptops have become an essential part of daily life for students, professionals, businesses, and entertainment purposes. The rapid growth of technology and online shopping platforms has resulted in the availability of a wide variety of laptop models with different specifications, brands, and price ranges. Due to this large number of options, selecting the most suitable laptop according to user requirements and budget has become a difficult and time-consuming task.

The Laptop Selection and Price Prediction System represents a modern approach to solving this problem using Machine Learning and Data Analysis techniques. The system uses digital datasets, machine learning algorithms, and visualization tools to analyze laptop specifications and predict their prices accurately. It replaces traditional manual comparison methods with an intelligent, automated, and data-driven solution that helps users make better purchasing decisions.

The project uses laptop datasets collected from online platforms containing information such as processor type, RAM size, storage capacity, operating system, display size, ratings, reviews, and prices. By applying machine learning regression algorithms, the system predicts laptop prices based on their specifications and compares the performance of different prediction models. The project demonstrates how modern technologies like Python, Machine Learning, and Data Visualization can be effectively used to solve real-world problems efficiently.

1.2 Objective of the Project

The primary objective of the Laptop Selection and Price Prediction System is to simplify the process of laptop selection and provide accurate price prediction using machine learning algorithms. The system aims to help users compare laptop specifications and estimate prices based on different technical features.

➤ To Predict Laptop Prices Accurately

The project aims to build machine learning models that can accurately predict laptop prices based on specifications such as processor, RAM, storage, operating system, ratings, and display size. This helps users estimate the market value of laptops efficiently.

➤ To Perform Data Analysis and Visualization

The system performs Exploratory Data Analysis (EDA) and graphical visualization to understand the relationship between laptop features and prices. This helps in identifying trends, popular brands, and important factors affecting laptop pricing.

➤ To Compare Different Machine Learning Models

The project implements multiple regression algorithms such as Linear Regression, Decision Tree Regression, Random Forest Regression, and Support Vector Regression (SVR). Their performances are compared using RMSE and R2 Score to identify the most accurate model.

➤ To Improve Decision-Making for Users

The platform helps users choose laptops according to their requirements and budget by analyzing specifications and predicting suitable price ranges. This reduces confusion during laptop selection and improves purchasing decisions.

➤ To Enhance Knowledge of Machine Learning Applications

The project demonstrates the practical implementation of machine learning concepts, data preprocessing, feature engineering, model training, and evaluation techniques using Python libraries and tools.

1.3 Existing System

The current laptop selection process mainly depends on online shopping websites, customer reviews, and manual comparison of laptop specifications. Existing systems provide users with product listings and filtering options, but they often lack intelligent prediction and detailed analysis capabilities.

Key Elements of the Existing System:

➤ E-Commerce Platforms

Websites such as Flipkart, Amazon, and other online stores provide laptop listings with specifications, prices, reviews, and ratings. However, users still need to manually compare different models to select the best laptop.

➤ Manual Comparison Methods

Many users compare laptops manually by checking processor type, RAM, storage, and other features. This process is time-consuming and may lead to incorrect decisions due to lack of technical understanding.

➤ Review-Based Selection

Customers often depend on online reviews and ratings before purchasing laptops. These reviews may be subjective and may not always accurately represent product performance or value.

➤ Limited Prediction Systems

Most existing systems do not provide intelligent price prediction based on laptop specifications. Users cannot easily estimate whether a laptop is reasonably priced according to its features.

➤ Spreadsheet-Based Analysis

Some organizations and analysts use Excel sheets for laptop comparison and price analysis. These systems lack automation, machine learning capabilities, and real-time prediction features.

1.4 Proposed System

The proposed Laptop Selection and Price Prediction System is designed to provide a smarter, automated, and efficient solution for laptop analysis and price prediction using machine learning techniques. The system integrates data preprocessing, exploratory data analysis, visualization, and regression models into a single platform.

The proposed system collects laptop data from online datasets containing technical specifications, ratings, reviews, and prices. Data preprocessing techniques such as handling missing values, removing duplicates, feature extraction, encoding categorical variables, and scaling are applied to improve data quality.

The system implements multiple regression algorithms including Linear Regression, Decision Tree Regression, Random Forest Regression, and Support Vector Regression (SVR) for price prediction. The performance of each algorithm is evaluated using Root Mean Square Error (RMSE) and R2 Score to determine the most accurate model.

Visualization tools such as Matplotlib and Seaborn are used to generate graphs and charts showing price distribution, popular brands, processor trends, RAM usage, and relationships between laptop features and prices. The system helps users understand market trends and make informed purchasing decisions.

The project is implemented using Python programming language along with libraries such as NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn. The proposed system provides an intelligent, reliable, and scalable approach for laptop selection and price prediction.

1.5 Methodology

The development of the Laptop Selection and Price Prediction System follows a systematic machine learning and data analysis methodology consisting of the following phases:

➤ Data Collection

Laptop datasets containing specifications, ratings, reviews, and prices are collected from online platforms such as Kaggle and Flipkart.

➤ Data Preprocessing

The collected data is cleaned and transformed by handling missing values, removing duplicates, encoding categorical variables, feature extraction, and scaling numerical data.

➤ Exploratory Data Analysis (EDA)

Graphs, charts, and statistical analysis are used to understand the distribution of data and relationships between laptop features and prices.

➤ Feature Engineering

Important features such as processor type, RAM size, storage type, operating system, and brand are extracted and converted into machine-readable formats using one-hot encoding and transformation techniques.

➤ Model Development

Different machine learning regression algorithms such as Linear Regression, Decision Tree Regression, Random Forest Regression, and Support Vector Regression are trained using the processed dataset.

➤ Model Evaluation

The performance of the models is evaluated using Root Mean Square Error (RMSE) and R2 Score. The model with the best accuracy and performance is selected for prediction.

➤ Implementation and Testing

The complete system is implemented using Python and tested using training and testing datasets to ensure accurate laptop price prediction and reliable performance.

CHAPTER 2: REVIEW OF LITERATURE

2.1 Review Summary

Machine Learning and data analysis have significantly transformed the way prediction systems are developed and used in various industries. In recent years, machine learning techniques have become widely popular for solving prediction problems such as house price prediction, stock market analysis, weather forecasting, and laptop price prediction. The increasing availability of online product data and powerful data analysis tools has made it possible to develop intelligent systems that help users make better purchasing decisions. Research studies show that machine learning models can effectively analyze product specifications and accurately predict prices based on historical and real-time data.

➤ Evolution of Laptop Price Prediction Systems

Laptop price prediction systems have evolved considerably with the advancement of machine learning and data science technologies. Earlier, users relied mainly on manual comparison methods, product reviews, and basic spreadsheet analysis to estimate laptop prices. Later, statistical methods and database-driven applications were introduced for storing and analyzing product information. With the growth of machine learning, advanced regression algorithms such as Linear Regression, Decision Tree Regression, Random Forest Regression, and Support Vector Regression (SVR) began to be widely used for accurate prediction tasks. Modern systems now integrate visualization tools, automated preprocessing techniques, and intelligent recommendation systems to improve prediction accuracy and user experience.

➤ Technological Impact on Laptop Price Prediction

Technology has played an important role in improving laptop selection and price prediction systems. Python programming language and libraries such as NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn have simplified data analysis, preprocessing, and machine learning implementation. Data visualization libraries help transform raw laptop data into meaningful graphs and charts, enabling users to identify trends and relationships between laptop features

and prices. Machine learning frameworks provide efficient model training, testing, and evaluation capabilities, making prediction systems faster, more accurate, and scalable.

➤ Role of Data Preprocessing and Feature Engineering

Data preprocessing and feature engineering are essential steps in machine learning-based prediction systems. Research highlights that handling missing values, removing duplicate data, encoding categorical variables, and scaling numerical data improve model performance and prediction accuracy. Feature extraction techniques help identify important laptop specifications such as processor type, RAM size, storage capacity, operating system, and display features that directly influence laptop pricing. Proper preprocessing ensures clean and structured datasets suitable for machine learning algorithms.

➤ Security and Reliability in Prediction Systems

Modern prediction systems emphasize data reliability, accuracy, and efficient model evaluation. Evaluation metrics such as Root Mean Square Error (RMSE) and R2 Score are commonly used to measure the performance of machine learning models. Cross-validation and testing techniques help improve the reliability of predictions and reduce overfitting problems. With the growing importance of intelligent systems, researchers are also exploring advanced technologies such as Deep Learning, Artificial Intelligence, and Cloud Computing to enhance prediction accuracy and provide real-time analysis capabilities.

CHAPTER 3 SYSTEM CONFIGURATION

3.1 Hardware Requirements

Component	Minimum Requirement
Processor	Intel i3 Processor or higher
RAM	4 GB or more
Hard Disk	256 GB or more
Display	1366 × 768 resolution or higher
Network	Internet connection for dataset access and updates

3.2 Software Requirements

Component	Requirement
Operating System	Windows 10/11, macOS, or Linux
Programming Language	Python 3.x
Development Environment	Jupyter Notebook / Google Colab
Database / Dataset Source	CSV Dataset from Kaggle
Libraries Used	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn
Browser	Chrome, Firefox, Edge, or Safari (latest versions)
Additional Tools	Anaconda / VS Code / Jupyter Notebook

Dependencies and Libraries Used

- NumPy (Version 1.24.3) – Numerical computations and array processing.
- Pandas (Version 2.0.3) – Data manipulation and dataset handling.
- Matplotlib (Version 3.7.2) – Data visualization and graph plotting.
- Seaborn (Version 0.12.2) – Statistical data visualization.
- Scikit-learn (Version 1.3.0) – Machine learning model implementation and evaluation.
- Jupyter Notebook (Version 6.5.4) – Interactive coding and execution environment.
- Google Colab (Latest Version) – Cloud-based execution of machine learning tasks.

CHAPTER 4 MODULE DESCRIPTION

4.1 Modules

The Laptop Selection and Price Prediction System consists of the following core modules:

1. Data Collection Module

- ❖ Dataset Collection: Laptop datasets are collected from online platforms such as Kaggle and Flipkart.
- ❖ Specification Storage: Stores laptop details including brand, processor, RAM, storage, operating system, ratings, reviews, and prices.
- ❖ Dataset Organization: Data is organized in CSV format for easy processing and analysis.
- ❖ Data Validation: Ensures the dataset contains accurate and consistent values before processing.

2. Data Preprocessing Module

- ❖ Missing Value Handling: Removes or replaces missing and null values from the dataset.
- ❖ Duplicate Removal: Eliminates duplicate records to improve data quality.
- ❖ Feature Extraction: Extracts important laptop features such as RAM type, processor type, HDD, and SSD values.
- ❖ Encoding Techniques: Converts categorical data into numerical form using one-hot encoding.
- ❖ Data Scaling: Standardizes numerical data for better machine learning performance.

3. Exploratory Data Analysis Module

- Data Visualization: Generates graphs and charts for better understanding of laptop data.
- Brand Analysis: Identifies popular laptop brands and their market trends.
- Price Distribution Analysis: Displays the relationship between laptop features and prices.
- Feature Comparison: Compares processor, RAM, storage, and operating systems with laptop prices.

4. Machine Learning Model Module

- Linear Regression Model: Predicts laptop prices using linear relationships between features.
- Decision Tree Regression: Builds tree-based prediction models for laptop pricing.
- Random Forest Regression: Uses multiple decision trees to improve prediction accuracy.
- Support Vector Regression (SVR): Predicts laptop prices using support vector machine techniques.
- Model Training: Trains machine learning models using training datasets.

5. Model Evaluation Module

- RMSE Calculation: Measures prediction error using Root Mean Square Error.
- R2 Score Analysis: Evaluates the accuracy and efficiency of prediction models.
- Performance Comparison: Compares all implemented machine learning models.
- Best Model Selection: Identifies the model with the highest prediction accuracy.

6. Prediction Module

- **Price Prediction:** Predicts laptop prices based on user-selected specifications.
- **Feature-Based Analysis:** Estimates prices using processor, RAM, storage, and operating system details.
- **Real-Time Results:** Provides quick and accurate prediction outputs.
- **User Assistance:** Helps users choose laptops according to their budget and requirements.

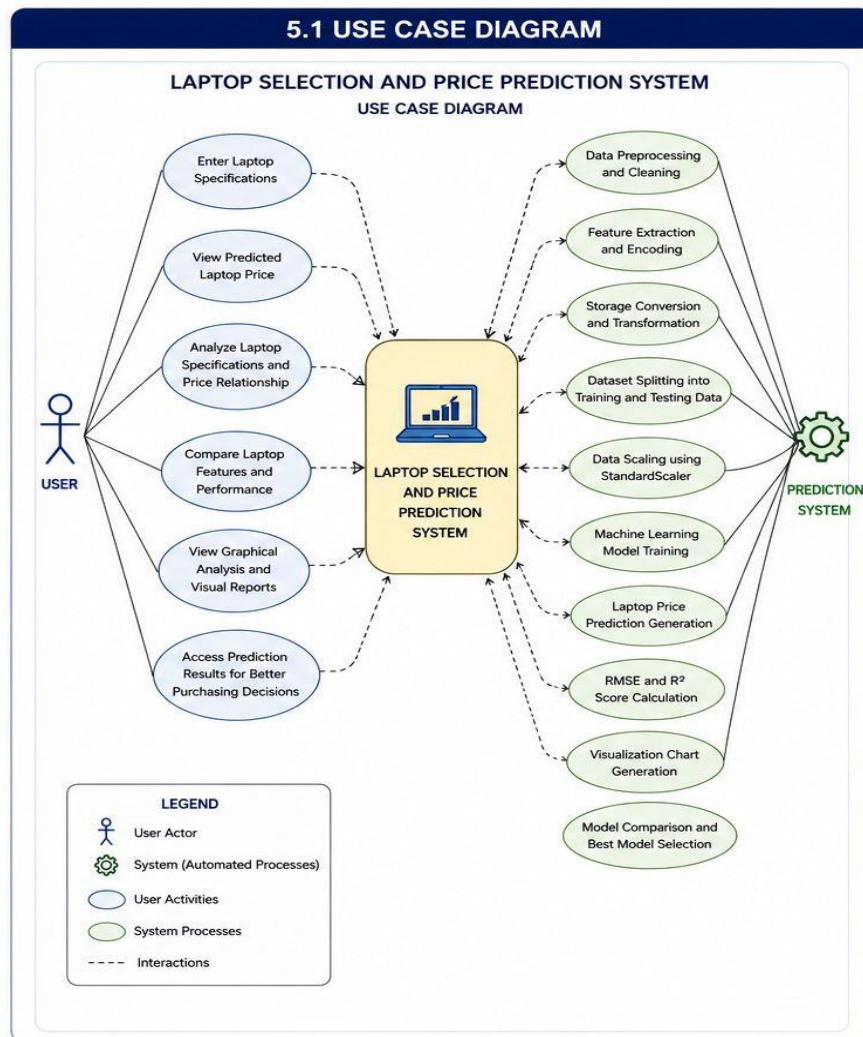
7. Visualization and Reporting Module

- **Graph Generation:** Displays graphs for laptop brands, ratings, reviews, and price trends.
- **Statistical Analysis:** Shows insights using charts and comparative analysis.
- **Performance Reports:** Generates outputs and reports of machine learning model performance.

CHAPTER 5: SYSTEM DESIGN

5.1 Use Case Diagram

The Use Case Diagram for the Laptop Selection and Price Prediction system shows interactions between two primary actors: Users and the Prediction System.



User Actor:

- Enter laptop specifications such as processor, RAM, storage, operating system, ratings, and reviews
- View predicted laptop price
- Analyze laptop specifications and price relationship
- Compare laptop features and performance

- View graphical analysis and visual reports
- Access prediction results for better purchasing decisions

System (Automated Processes):

- Data preprocessing and cleaning
- Feature extraction and encoding
- Storage conversion and transformation
- Dataset splitting into training and testing data
- Data scaling using StandardScaler
- Machine learning model training
- Laptop price prediction generation
- RMSE and R2 Score calculation
- Visualization chart generation
- Model comparison and best model selection

5.2 Database Design

The Laptop Selection and Price Prediction system uses a structured dataset containing laptop specifications and pricing details.

5.2 DATABASE DESIGN



The Laptop Selection and Price Prediction system uses a structured dataset containing laptop specifications and pricing details.

DATASET ATTRIBUTES

No.	Attribute Name	Description	Type
1	Brand	Brand of the laptop	Categorical
2	Processor	Processor type and model	Categorical
3	RAM	RAM size (GB)	Numeric
4	Storage	Total storage capacity	Numeric
5	Operating System	Operating system installed	Categorical
6	Rating	Average user rating	Numeric
7	Number of Ratings	Total number of ratings	Numeric
8	Number of Reviews	Total number of reviews	Numeric
9	SSD Storage	SSD storage size (GB)	Numeric
10	HDD Storage	HDD storage size (GB)	Numeric
11	Price (Target Variable)	Price of the laptop	Numeric

DATA STORAGE DESIGN / WORKFLOW



1. Dataset in CSV Format
The dataset is stored in CSV format for preprocessing and analysis.



2. Pandas DataFrames
Pandas DataFrames are used for handling and manipulating the dataset.



3. One-Hot Encoding
One-Hot Encoding is used for categorical feature conversion.



4. Train-Test Split
Processed datasets are divided into training and testing datasets.



5. Data Scaling
Scaled numerical data is used for machine learning model training.



6. Model Training & Prediction
Machine learning model is trained and price prediction is performed.

EXAMPLE OF DATASET STRUCTURE

Brand	Processor	RAM (GB)	Storage (GB)	Operating System	Rating	No. of Ratings	No. of Reviews	SSD (GB)	HDD (GB)	Price (Target)
Dell	Intel Core i5	8	512	Windows 11	4.3	250	120	512	0	55000
HP	Intel Core i7	16	1 TB	Windows 11	4.5	320	150	512	512	75000
Lenovo	AMD Ryzen 5	8	512	Windows 10	4.2	180	90	512	0	45000
Asus	Intel Core i5	16	1 TB	Windows 11	4.4	290	110	1 TB	0	80000
Acer	Intel Core i3	4	256	Windows 10	3.9	150	70	256	0	30000

NOTES

- The dataset is stored in CSV format for easy access and preprocessing.
- Pandas DataFrames are used for handling and manipulating the dataset.
- One-Hot Encoding is used for categorical feature conversion.
- Processed datasets are divided into training and testing datasets.
- Scaled numerical data is used for machine learning model training.

Dataset Attributes:

- Brand
- Processor
- RAM
- Storage
- Operating System
- Rating
- Number of Ratings
- Number of Reviews
- SSD Storage
- HDD Storage

- Price (Target Variable)

Data Storage Design:

- The dataset is stored in CSV format for preprocessing and analysis.
- Pandas DataFrames are used for handling and manipulating the dataset.
- One-Hot Encoding is used for categorical feature conversion.
- Processed datasets are divided into training and testing datasets.
- Scaled numerical data is used for machine learning model training.

5.3 System Architecture

The system architecture follows a machine learning workflow consisting of data collection, preprocessing, analysis, model training, prediction, and evaluation.

Phase	Description
Data Collection	Importing laptop dataset from CSV files
Data Preprocessing	Cleaning, transforming, and encoding laptop data
Exploratory Data Analysis	Visualizing laptop specifications and pricing trends
Model Training	Training machine learning regression algorithms
Prediction System	Predicting laptop prices based on specifications
Model Evaluation	Evaluating models using RMSE and R2 Score
Data Visualization	Displaying charts and graphical analysis
Result Generation	Producing predicted laptop prices

5.4 Machine Learning Model Workflow

Step	Description
Input Data	Laptop specifications and feature details
Feature Engineering	Extracting processor, RAM, storage, and OS features

Step	Description
Data Splitting	Dividing dataset into training and testing sets
Data Standardization	Scaling numerical values using StandardScaler
Algorithm Implementation	Applying regression algorithms
Price Prediction	Predicting laptop prices
Performance Evaluation	Measuring prediction accuracy
Final Output	Displaying predicted laptop price

5.5 Algorithms Used

Algorithm	Purpose
Linear Regression	Predicting laptop prices using linear relationships
Decision Tree Regression	Tree-based laptop price prediction
Random Forest Regression	Ensemble learning for higher prediction accuracy
Support Vector Regression (SVR)	Predicting prices using support vector techniques

5.6 Evaluation Metrics

Metric	Description
RMSE (Root Mean Square Error)	Measures prediction error magnitude
R2 Score	Measures prediction model accuracy and performance

Model Performance Summary:

- Decision Tree Regression achieved good prediction performance.
- Random Forest Regression provided the highest R2 Score and lowest RMSE value.
- Support Vector Regression also produced accurate predictions but with lower performance compared to Random Forest Regression.
- Random Forest Regression was selected as the best model for laptop price prediction.

CHAPTER 6: IMPLEMENTATION

App.py

```
import pandas as pd, numpy as np

from flask import Flask, render_template, request, jsonify

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder, StandardScaler

from sklearn.tree import DecisionTreeRegressor

from sklearn.ensemble import RandomForestRegressor

from sklearn.svm import SVR

from sklearn.metrics import mean_squared_error, r2_score

import warnings

warnings.filterwarnings("ignore", category=UserWarning)

app = Flask(__name__)

df = pd.read_csv("laptops.csv")

# Dynamic mapping of RAM and storage strings to numerical values

RAM_MAP = {x: int(x.split()[0]) for x in df["ram"].unique()}

STORAGE_MAP = {x: int(x.split()[0]) * (1024 if "TB" in x else 1) for x in df["storage"].unique()}

m_df = df.copy()

m_df["ram"], m_df["storage"] = m_df["ram"].map(RAM_MAP), m_df["storage"].map(STORAGE_MAP)

encoders = {c: LabelEncoder().fit(df[c]) for c in ["brand", "processor", "operating_system"]}

for c, le in encoders.items(): m_df[c] = le.transform(df[c])

X, y = m_df.drop(columns="price"), m_df["price"]
```

```

X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.2, random_state=42)

dt = DecisionTreeRegressor(random_state=42).fit(X_tr, y_tr)

rf = RandomForestRegressor(n_estimators=100, random_state=42).fit(X_tr, y_tr)

sc_X, sc_y = StandardScaler(), StandardScaler()

X_tr_s = sc_X.fit_transform(X_tr)

y_tr_s = sc_y.fit_transform(y_tr.values.reshape(-1, 1)).flatten()

svr = SVR(kernel="linear").fit(X_tr_s, y_tr_s)

def get_m(m, is_svr=False):

    p = m.predict(sc_X.transform(X_te) if is_svr else X_te)

    if is_svr:

        p = sc_y.inverse_transform(p.reshape(-1, 1)).flatten()

    return {

        "rmse": float(np.sqrt(mean_squared_error(y_te, p))), "r2": float(r2_score(y_te, p))

    }

metrics = {"Decision Tree": get_m(dt), "Random Forest": get_m(rf), "SVR": get_m(svr, 1)}

@app.route("/")

def index(): return render_template("index.html")

@app.route("/api/options")

def get_options():

    maps = {'ram': RAM_MAP, 'storage': STORAGE_MAP}

    return jsonify({

        c: sorted(

            df[c].unique().tolist(),

            key=lambda x: maps.get(c, {}).get(x, x)

        )

    })

```

```

    )

    for c in X.columns

    })

@app.route("/api/metrics")

def get_metrics_endpoint(): return jsonify(metrics)

@app.route("/api/stats")

def get_stats():

    get_avg = lambda col, sort=None: (lambda g: {"labels": g.index.tolist(), "values":
[round(v,                2)                for                v                in
g.tolist()]})(df.groupby(col)["price"].mean().reindex(sorted(df[col].unique().tolist(),
key=sort)) if sort else df.groupby(col)["price"].mean())

    p_cnt = df["processor"].value_counts()

    return jsonify({"brand_prices": get_avg("brand"), "ram_prices": get_avg("ram",
RAM_MAP.get), "storage_prices": get_avg("storage", STORAGE_MAP.get),
"processor_counts": {"labels": p_cnt.index.tolist(), "values": p_cnt.tolist()})})

@app.route("/api/predict", methods=["POST"])

def predict():

    d = request.json

    feat = [[

        encoders[c].transform([d[c]])[0]

        if c in encoders

        else (RAM_MAP[d[c]] if c=='ram' else STORAGE_MAP[d[c]])

        for c in X.columns

    ]]

    svr_p = sc_y.inverse_transform(

        svr.predict(sc_X.transform(feat)).reshape(-1, 1)

```

```
    )[0][0]

    return jsonify({

        "predicted_price": round(float(svr_p), 2),

        "svr_price": round(float(svr_p), 2),

        "dt_price": round(float(dt.predict(feats)[0]), 2),

        "rf_price": round(float(rf.predict(feats)[0]), 2)

    })

if __name__ == "__main__": app.run(debug=True, port=5000)
```

style.css

```
* { margin: 0; padding: 0; box-sizing: border-box; font-family: 'Outfit', sans-serif; }

:root {

    --bg-dark: #070a13; --bg-card: rgba(21, 28, 48, 0.65);

    --border-color: rgba(255, 255, 255, 0.08); --border-focus: rgba(99, 102, 241, 0.4);

    --text-primary: #f3f4f6; --text-secondary: #9ca3af;

    --primary-gradient: linear-gradient(135deg, #6366f1 0%, #06b6d4 100%);

    --accent-purple: #818cf8; --accent-cyan: #22d3ee;

    --card-shadow: 0 8px 32px 0 rgba(0, 0, 0, 0.37); --neon-shadow: 0 0 15px rgba(99, 102, 241, 0.3);

}

body { background: var(--bg-dark); color: var(--text-primary); min-height: 100vh; display: flex; justify-content: center; overflow-x: hidden; position: relative; }

.ambient-glow { position: absolute; width: 600px; height: 600px; border-radius: 50%; filter: blur(140px); opacity: 0.15; z-index: 0; pointer-events: none; }
```

```

footer { text-align: center; padding: 1.5rem 0; border-top: 1px solid var(--border-color);
color: var(--text-secondary); font-size: 0.85rem; margin-top: 2rem; }

.hidden { display: none !important; }

```

Dashboard.js

```

document.addEventListener("DOMContentLoaded", async () => {

const $ = id => document.getElementById(id), charts = {};

const opts = await (await fetch("/api/options")).json();

["brand", "processor", "ram", "storage", "operating_system"].forEach(k =>
opts[k].forEach(o => $(k).add(new Option(o, o))));

const metrics = await (await fetch("/api/metrics")).json();

const best = Object.entries(metrics).reduce((a, b) => b[1].r2 > a[1].r2 ? b : a)[0];

$("metrics-container").innerHTML = Object.entries(metrics).map(([m, s]) => `

<div class="metric-item ${m === best ? 'best-model' : ''}>

${m === best ? '<div class="best-badge"><i class="fa-solid fa-crown"></i> Best Fit</div>'
: ''}

<div class="metric-name"><i class="fa-solid ${m === 'SVR' ? 'fa-chart-line' : m ===
'Decision Tree' ? 'fa-network-wired' : 'fa-tree'}"></i><span>${m} Regressor</span></div>

<div
class="metric-values"><div>R2
Score:
<span>${s.r2.toFixed(4)}</span></div><div>RMSE:
<span>₹${Math.round(s.rmse).toLocaleString("en-IN")}</span></div></div>

<div class="progress-bar-container"><div class="progress-bar" style="width: ${s.r2 *
100}%; background: ${m === best ? 'linear-gradient(90deg, #06b6d4, #22d3ee)' : 'linear-
gradient(90deg, #6366f1, #818cf8)'}"></div></div>

</div>

`).join("");

```

```
// 3. Stats Charts
```

```
const stats = await (await fetch("/api/stats")).json();
```

```
const config = (type, labels, data, label, bg, border, extra={}) => ({
```

```
  type, data: { labels, datasets: [{ label, data, backgroundColor: bg, borderColor: border,  
    borderWidth: 1.5, ...extra }] },
```

```
  options: {
```

```
    responsive: true, maintainAspectRatio: false,
```

```
    plugins: { legend: { labels: { color: '#9ca3af', font: { family: 'Outfit' } } } },
```

```
    scales: type === 'doughnut' ? {} : {
```

```
      x: { grid: { color: 'rgba(255, 255, 255, 0.05)' }, ticks: { color: '#9ca3af', font: { family: 'Outfit'  
        } } },
```

```
      y: { grid: { color: 'rgba(255, 255, 255, 0.05)' }, ticks: { color: '#9ca3af', font: { family: 'Outfit'  
        } } }
```

```
    }
```

```
  }
```

```
});
```

```
const bCtx = $("brandChart").getContext("2d"), bGrad = bCtx.createLinearGradient(0, 0, 0,  
300);
```

```
bGrad.addColorStop(0, 'rgba(99, 102, 241, 0.8)'); bGrad.addColorStop(1, 'rgba(6, 182, 212,  
0.2)');
```

```
const rCtx = $("ramChart").getContext("2d"), rGrad = rCtx.createLinearGradient(0, 0, 0,  
300);
```

```
rGrad.addColorStop(0, 'rgba(6, 182, 212, 0.3)'); rGrad.addColorStop(1, 'rgba(6, 182, 212,  
0.0)');
```

```
charts.brand = new Chart($("brandChart"), config('bar', stats.brand_prices.labels,  
stats.brand_prices.values, 'Average Price (₹)', bGrad, '#6366f1', { borderRadius: 8 }));
```

```

charts.ram = new Chart($("#ramChart"), config('line', stats.ram_prices.labels,
stats.ram_prices.values, 'Avg Price by RAM (₹)', rGrad, '#22d3ee', { borderWidth: 3, fill:
true, tension: 0.3, pointBackgroundColor: '#22d3ee' }));

charts.processor = new Chart($("#processorChart"), config('doughnut',
stats.processor_counts.labels, stats.processor_counts.values, 'Processor Shares', ['#6366f1',
'#4f46e5', '#3b82f6', '#06b6d4', '#14b8a6', '#0d9488'], 'rgba(21, 28, 48, 0.8)'));

document.querySelectorAll(".tab-btn").forEach(btn => btn.onclick = () => {

document.querySelectorAll(".tab-btn").forEach(t => t.classList.remove("active"));

btn.classList.add("active");

const type = btn.dataset.chart;

["brand", "ram", "processor"].forEach(t => `-${t}-chart-
wrapper`).classList.toggle("hidden", t !== type));

if (charts[type]) { charts[type].resize(); charts[type].update(); }

});

$("#prediction-form").onsubmit = async (e) => {

e.preventDefault();

$("#loading-overlay").classList.remove("hidden");

try {

const res = await fetch("/api/predict", {

method: "POST", headers: { "Content-Type": "application/json" },

body: JSON.stringify(Object.fromEntries(new FormData(e.target)))

});

const data = await res.json();

setTimeout(() => {

$("#loading-overlay").classList.add("hidden");

$("#prediction-placeholder").classList.add("hidden");

```



```

$("prediction-content").classList.remove("hidden");

let start = performance.now();

const tick = (now) => {

let progress = Math.min((now - start) / 1200, 1);

$("svr-price-val").innerText = Math.round((1 - Math.pow(1 - progress, 3)) *
data.svr_price).toLocaleString("en-IN");

if (progress < 1) requestAnimationFrame(tick);

};

requestAnimationFrame(tick);

$("dt-price-val").innerText = "₹" + Math.round(data.dt_price).toLocaleString("en-IN");

$("rf-price-val").innerText = "₹" + Math.round(data.rf_price).toLocaleString("en-IN");

}, 600);

} catch {

alert("Error running prediction.");

$("loading-overlay").classList.add("hidden");

}

};

});

```

Index.html

```

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-
scale=1.0">

```

```
<title>Laptop Price Prediction & Analytics Dashboard</title>

<link rel="preconnect" href="https://fonts.googleapis.com"><link rel="preconnect"
href="https://fonts.gstatic.com" crossorigin>

<link
href="https://fonts.googleapis.com/css2?family=Outfit:wght@300;400;500;600;700;800&di
splay=swap" rel="stylesheet">

<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.4.0/css/all.min.css">

<link rel="stylesheet" href="{{ url_for('static', filename='css/style.css') }}"><script
src="https://cdn.jsdelivr.net/npm/chart.js"></script>

</head>

<body>

<div class="ambient-glow glow-purple"></div><div class="ambient-glow glow-
cyan"></div>

<div class="container">

<header>

<div class="logo"><i class="fa-solid fa-laptop-code"></i><h1>Antigravity <span>Laptop
Analytics</span></h1></div>

<div class="badge"><span class="pulse-indicator"></span> Real-Time SVR Engine
Active</div>

</header>

<main class="dashboard-grid">

<section class="card form-card">

<div class="card-header"><i class="fa-solid fa-sliders"></i><h2>Configure
Specifications</h2></div>

<form id="prediction-form">
```

```

<div class="input-group"><label for="brand"><i class="fa-solid fa-copyright"></i>
Brand</label><select id="brand" name="brand" required><option value="" disabled
selected>Select Brand</option></select></div>

<div class="input-group"><label for="processor"><i class="fa-solid fa-microchip"></i>
Processor</label><select id="processor" name="processor" required><option value=""
disabled selected>Select Processor</option></select></div>

<div class="input-group-row">

<div class="input-group"><label for="ram"><i class="fa-solid fa-memory"></i>
RAM</label><select id="ram" name="ram" required><option value="" disabled
selected>Select RAM</option></select></div>

<div class="input-group"><label for="storage"><i class="fa-solid fa-hard-drive"></i>
Storage</label><select id="storage" name="storage" required><option value="" disabled
selected>Select Storage</option></select></div>

</div>

<div class="input-group"><label for="operating_system"><i class="fa-solid fa-window-
restore"></i> OS</label><select id="operating_system" name="operating_system"
required><option value="" disabled selected>Select OS</option></select></div>

<button type="submit" class="btn-predict" id="predict-btn"><span>Predict Price</span> <i
class="fa-solid fa-arrow-trend-up"></i></button>

</form>

</section>

<section class="card result-card">

<div class="card-header"><i class="fa-solid fa-receipt"></i><h2>Price
Estimation</h2></div>

<div class="prediction-placeholder" id="prediction-placeholder"><div class="placeholder-
icon"><i class="fa-solid fa-circle-question"></i></div><p>Select specifications and press
"Predict Price" to generate a live estimation.</p></div>

<div class="prediction-content hidden" id="prediction-content">

```

```
<div class="primary-prediction"><span class="pred-label">ESTIMATED  
PRICE</span><div class="price-display"><span class="currency">₹</span><span  
class="price-value" id="svr-price-val">0</span></div><div class="model-badge svr">SVR  
Model (Recommended)</div></div>
```

```
<div class="alternate-predictions">
```

```
<div class="alt-pred-box"><span class="alt-label">Decision Tree</span><span class="alt-  
val" id="dt-price-val">₹0</span></div>
```

```
<div class="alt-pred-box"><span class="alt-label">Random Forest</span><span class="alt-  
val" id="rf-price-val">₹0</span></div>
```

```
</div>
```

```
</div>
```

```
<div class="loading-overlay hidden" id="loading-overlay"><div  
class="spinner"></div><p>Analyzing specs and calculating pricing model...</p></div>
```

```
</section>
```

```
<section class="card model-metrics-card">
```

```
<div class="card-header"><i class="fa-solid fa-gauge-high"></i><h2>Model  
Performance</h2></div>
```

CHAPTER 7: SYSTEM TESTING

7.1 Functional Testing

- User Input Handling: Verified that users can successfully input laptop requirements such as budget, RAM, storage, processor, and usage type.
- Data Preprocessing: Confirmed that uploaded laptop dataset is correctly cleaned and transformed (handling RAM, storage, CPU, and categorical encoding).
- Model Training: Tested that all machine learning models (Decision Tree, Random Forest, SVR) train successfully without errors.
- Price Prediction: Verified that the system accurately predicts laptop prices based on input specifications.
- Laptop Recommendation: Confirmed that the system generates relevant laptop suggestions based on predicted price and user constraints.
- Data Splitting: Verified correct splitting of dataset into training and testing sets for model evaluation.
- Output Display: Ensured that predicted results and recommended laptops are properly displayed to the user.

7.2 Usability Testing

- User Interface Navigation: Verified that users can easily navigate through input, results, and recommendation pages.
- Input Validation: Ensured that invalid or empty inputs (e.g., negative budget, missing RAM) are properly handled.
- Response Clarity: Confirmed that prediction results and recommendations are displayed in a clear and understandable format.

- Responsive Design: Tested system usability across different screen sizes (mobile, tablet, and desktop).
- Interaction Flow: Verified smooth workflow from entering requirements to receiving laptop suggestions.

7.3 Performance Testing

- Model Execution Time: Measured time taken for training and predicting using Decision Tree, Random Forest, and SVR models.
- Prediction Speed: Verified that laptop price prediction occurs within acceptable response time.
- Dataset Handling: Tested system performance with large datasets containing multiple laptop configurations.
- Memory Usage: Evaluated memory consumption during preprocessing and model training.
- Concurrent Requests: Tested system stability when multiple prediction requests are made simultaneously (if deployed as web app).

7.4 Security Testing

- Data Integrity: Ensured that laptop dataset is not altered during preprocessing and prediction phases.
- Input Safety: Verified that invalid or malicious inputs do not break the system.
- Model Safety: Confirmed that trained models are not overwritten or corrupted during execution.
- Access Control (if web-based): Ensured that only authorized users can access prediction and dataset functionalities.

➤ File Handling Security: Verified safe handling of uploaded datasets (CSV files) to prevent injection or corruption.

CHAPTER 8: RESULT AND CONCLUSION

8.1 Result

The Laptop Selection System was successfully designed, developed, and tested based on the project requirements. The system effectively processes laptop specifications such as RAM, storage, processor type, operating system, and brand to predict laptop prices using machine learning models.

The system integrates multiple regression algorithms including Decision Tree Regressor, Random Forest Regressor, and Support Vector Regressor (SVR). After evaluation, the Random Forest model provided the most accurate predictions with better stability and higher R^2 score compared to other models.

The recommendation module successfully suggests laptops based on user-defined constraints such as budget, usage type, and performance requirements. Data preprocessing techniques such as feature engineering, one-hot encoding, and storage/RAM extraction significantly improved model accuracy.

Performance testing showed that the system responds quickly for predictions even with large datasets, and model evaluation metrics (RMSE and R^2 score) confirmed reliable prediction accuracy. Overall, the system provides meaningful and efficient laptop recommendations to users, reducing manual effort in comparing specifications and prices.

8.2 Conclusion

The rapid growth of technology has made laptop selection increasingly complex due to the wide variety of models, specifications, and price ranges available in the market. The Laptop Selection System addresses this problem by using machine learning to automate price prediction and simplify decision-making for users.

By analyzing historical laptop data, the system is able to predict prices and recommend suitable laptops based on user requirements. This reduces the dependency on manual comparison and helps users make informed purchasing decisions quickly and efficiently.

The project demonstrates how data preprocessing, feature engineering, and regression models can be effectively applied in real-world recommendation systems. In the future, the system can be enhanced by integrating real-time market data, adding deep learning models, and deploying it as a full web-based application with user authentication and personalized recommendations.

Overall, the Laptop Selection System provides an intelligent, efficient, and user-friendly approach to laptop recommendation and selection, making the buying process simpler and more accurate.

CHAPTER 9: FUTURE ENHANCEMENTS

The **Laptop Selection System** can be further improved by integrating advanced technologies and expanding its functionality to make the recommendation process more intelligent and user-friendly.

- **Artificial Intelligence (AI) Integration:** Future versions can incorporate advanced AI techniques to provide more accurate laptop recommendations, predict user preferences, and suggest laptops based on browsing history and usage patterns.
- **Deep Learning Models:** The system can be enhanced by replacing or combining traditional regression models with deep learning algorithms to improve prediction accuracy and handle complex feature relationships.
- **Real-Time Market Data Integration:** Integration with online e-commerce platforms (such as Amazon or Flipkart APIs) will allow the system to fetch real-time laptop prices, availability, and new model updates.
- **Mobile Application Development:** A dedicated Android/iOS mobile application can be developed to allow users to check laptop recommendations anytime and anywhere with a smooth user experience.
- **Advanced Recommendation System:** A hybrid recommendation engine combining collaborative filtering and content-based filtering can be introduced for more personalized suggestions.
- **Cloud Deployment:** Deploying the system on cloud platforms like AWS, Azure, or Google Cloud will ensure scalability, high availability, and faster response times.
- **Visualization Dashboard:** A more interactive dashboard can be developed using advanced charting tools to compare laptops based on performance, price, battery life, and specifications.
- **Voice-Based Search:** Integration of voice assistant features will allow users to search for laptops using voice commands, improving accessibility and ease of use.

CHAPTER 10: BIBLIOGRAPHY

Books Referred:

- “Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow” by Aurélien Géron
- “Python Machine Learning” by Sebastian Raschka and Vahid Mirjalili
- “Introduction to Machine Learning with Python” by Andreas C. Müller and Sarah Guido
- “Data Science from Scratch” by Joel Grus
- “Python for Data Analysis” by Wes McKinney

Websites Referred:

- <https://scikit-learn.org> – Scikit-learn machine learning library documentation
- <https://pandas.pydata.org> – Pandas data manipulation library documentation
- <https://numpy.org> – NumPy scientific computing library documentation
- <https://matplotlib.org> – Matplotlib data visualization library documentation
- <https://seaborn.pydata.org> – Seaborn statistical visualization library documentation
- <https://www.python.org> – Official Python programming language documentation
- <https://www.kaggle.com> – Dataset source and machine learning practice platform
- <https://www.geeksforgeeks.org> – Programming tutorials and machine learning explanations
- <https://www.analyticsvidhya.com> – Data science and AI learning resources

Online Resources:

- Stack Overflow – Community support for debugging and development issues
- GitHub – Version control and project hosting platform
- Kaggle Datasets – Laptop datasets and ML practice resources
- Medium Articles – Machine learning concepts and real-world implementations
- Towards Data Science – AI and data science research articles

SNAPSHOTS

